ESOFT METRO CAMPUS

LONDON METROPOLITAN UNIVERSITY

Project details

Title	: AI-Powered Facial Expression Analysis System to Support Autism Screening with an Augmentative and Alternative Communication Platform for Children in Sri Lanka
Student Name	: Elabada Kankanamge Yasas Pasindu Fernando
Registration No	: E280041
User ID	:EKF002
Supervisor	: Ms. Niruni Fonseka
Second Supervisor	: [leave blank]
Date Submitted	: 2026/01/12
Date Approved	: [leave blank]

ACKNOWLEDGMENTS

The author wishes to express sincere gratitude to Ms. Niruni Fonseka for her continuous guidance, constructive feedback, and academic supervision throughout the development of this project proposal. Her direction has been invaluable in shaping both the research and practical aspects of the study.

The author also acknowledges the Ministry of Health, Sri Lanka, for recognizing the importance of research that aims to support children with special needs and contribute to socially beneficial technological solutions within the healthcare sector.

Special appreciation is extended to Karapitiya Teaching Hospital and its medical professionals for providing expert consultation and clinical insight related to autism screening practices. Their guidance ensures that the proposed system remains ethically sound, clinically relevant, and aligned with real-world healthcare requirements.

Finally, the author gratefully acknowledges Esoft Metro Campus for providing the academic framework, resources, and learning environment necessary to pursue this research project.

Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects social interaction, communication, and behaviour. According to global estimates, approximately one in 160 children worldwide is diagnosed with autism, highlighting the growing need for early identification and intervention *(World Health Organization, 2021)*. In Sri Lanka, access to timely autism screening and effective communication support remains limited due to resource constraints, high costs, and the lack of localized technological solutions.

This project proposes the development of an AI-powered facial expression analysis system designed to support autism screening by identifying facial patterns and expressions associated with autism. The system is intended as a preliminary screening support tool for medical professionals and does not replace formal clinical diagnosis. In parallel, a multilingual Augmentative and Alternative Communication (AAC) platform is proposed to assist children with autism in expressing their needs and emotions in Sinhala, Tamil, and English.

By combining artificial intelligence, computer vision, and mobile application technologies, the proposed solution aims to address both a research challenge and a social need. With expert consultation from Karapitiya Teaching Hospital and alignment with national healthcare priorities, the project seeks to contribute to more accessible, affordable, and culturally appropriate support for children with autism in Sri Lanka.

Background and Motivation

Recent studies indicate that facial expression recognition using machine learning techniques can support autism screening with high accuracy under controlled research conditions *(Samad et al., 2021)*. These findings highlight the potential of AI-assisted screening tools to reduce the burden on specialists and improve early referral processes.

Following diagnosis, many children with autism experience difficulties in expressing emotions and needs verbally. AAC systems are widely recognised as effective tools for supporting communication development and do not hinder natural speech acquisition *(Millar et al., 2006)*. However, most commercially available AAC solutions are expensive and do not support Sinhala and Tamil languages, making them inaccessible to many families in Sri Lanka.

During the December annual leave period of the previous year, the author had the opportunity to engage in discussions with friends affiliated with the Faculty of Medicine, Karapitiya. Through these discussions, it became evident that many families in Sri Lanka are unable to afford existing assistive communication applications due to their high cost, and that locally available solutions lack advanced AI-based features. These interactions highlighted a significant gap between the technological solutions available globally and their accessibility for children from low-income families in Sri Lanka.

Motivated by this realization, the author developed an initial prototype of the proposed application and presented it to medical professionals at Karapitiya Teaching Hospital. The prototype was positively received, and the medical team expressed strong interest in the system, emphasizing the urgent need for affordable assistive technologies for children with autism. Following this response, the author formally informed the project supervisor and decided to pursue the development of this system as a research project. Approval and encouragement were also obtained from the Ministry of Education to proceed with the initiative, and Karapitiya Teaching Hospital agreed to provide full professional support in the form of expert consultation and validation guidance. The author possesses strong technical skills in mobile application development, artificial intelligence, and user interface design, and is genuinely committed to applying these skills to develop an accessible and affordable assistive communication platform for children with autism.

Problem in Brief

Sri Lankan healthcare institutions face challenges in conducting early autism screening due to limited specialist availability and time-intensive assessment procedures. There is currently no AI-based facial expression analysis system adapted for local research and clinical support contexts.

In addition, children diagnosed with autism lack access to affordable and culturally appropriate AAC communication tools. Existing solutions are costly and do not support Sinhala and Tamil languages, creating communication barriers for children and increasing stress for families.

Aim

The aim of this project is to develop an AI-powered facial expression analysis system to support autism screening together with a multilingual Augmentative and Alternative Communication platform for children with autism, making use of Flutter, TensorFlow, OpenCV, Firebase, and mobile application technologies.

Objectives

1. To conduct a critical review of existing autism screening approaches and Augmentative and Alternative Communication systems.

2. To investigate suitable machine learning and computer vision techniques for facial expression analysis.

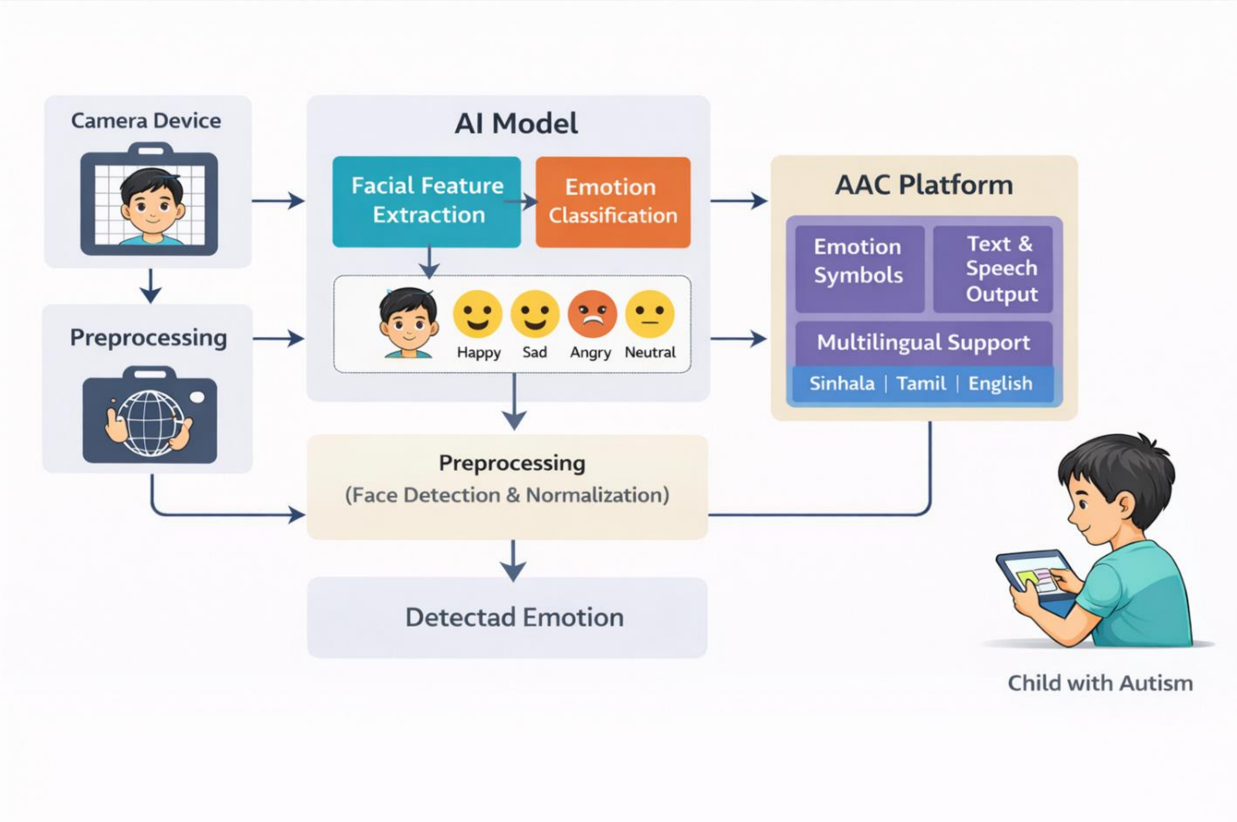
3. To design and develop an AI-powered facial expression analysis system using publicly available datasets.

4. To design and implement a multilingual AAC communication platform for children with autism.

5. To evaluate the effectiveness and usability of the proposed system through expert consultation and user feedback.

6. To document the system design, implementation, evaluation results, and ethical considerations.

Proposed Solution



The proposed solution consists of an integrated system designed to address two major challenges faced by children with autism in Sri Lanka: the lack of accessible autism screening support and the absence of affordable communication tools for low-income families.

The first component of the system is an AI-powered facial expression analysis module developed as a research-focused screening support tool. Facial images captured through a camera-enabled device are processed using computer vision techniques, including face detection, alignment, and normalization. Facial landmarks and expression-related features are then extracted and analysed using a convolutional neural network model trained on publicly available academic datasets. The trained model identifies facial expression patterns that may be associated with autism-related behavioural characteristics. The output of this system is intended to support medical professionals as a preliminary screening aid and does not replace clinical diagnosis. Expert validation guidance is provided by medical professionals from Karapitiya Teaching Hospital to ensure clinical relevance and ethical compliance.

The second component of the system is a multilingual Augmentative and Alternative Communication platform developed as a social support solution for children with autism, particularly those from low-income families who cannot afford existing commercial AAC applications. The AAC platform enables children to express their needs, emotions, and ideas using visual symbols, images, text, and speech output. The application supports Sinhala, Tamil, and English languages, allowing children to communicate naturally within their home and school environments. The platform is designed as an adaptive communication tool that can be customized according to each child's individual abilities and preferences and can operate offline for rural and low-resource environments.

Resource Requirements

The project requires standard development hardware, including a high-performance laptop and mobile test devices, along with software tools such as Flutter, TensorFlow, Firebase, and design tools for UI development. Publicly available datasets will be used for model training, and expert consultation will be obtained from medical professionals for validation purposes.

Hardware Requirements

- Development Laptop (i7 processor, 16GB RAM, GPU)
- Android and iOS Test Devices
- High-resolution Camera for test image capture
- Cloud GPU services for model training

Software Requirements

- Flutter SDK
- Android Studio
- Visual Studio Code
- TensorFlow and TensorFlow Lite
- OpenCV
- Firebase (Realtime Database, Cloud Functions)
- Figma / Photoshop for UI design

Deliverables

The project will deliver a free and accessible AI-powered facial expression analysis mobile application designed to support preliminary autism screening, together with a multilingual Augmentative and Alternative Communication platform for children with autism. The communication application will support Sinhala, Tamil, and English languages and will be made available at no cost in order to benefit children from low-income families. In addition, a trained machine learning model optimized for mobile deployment will be produced. A comprehensive research report, system documentation, and user manuals for parents and healthcare professionals will also be delivered at the completion of the project.

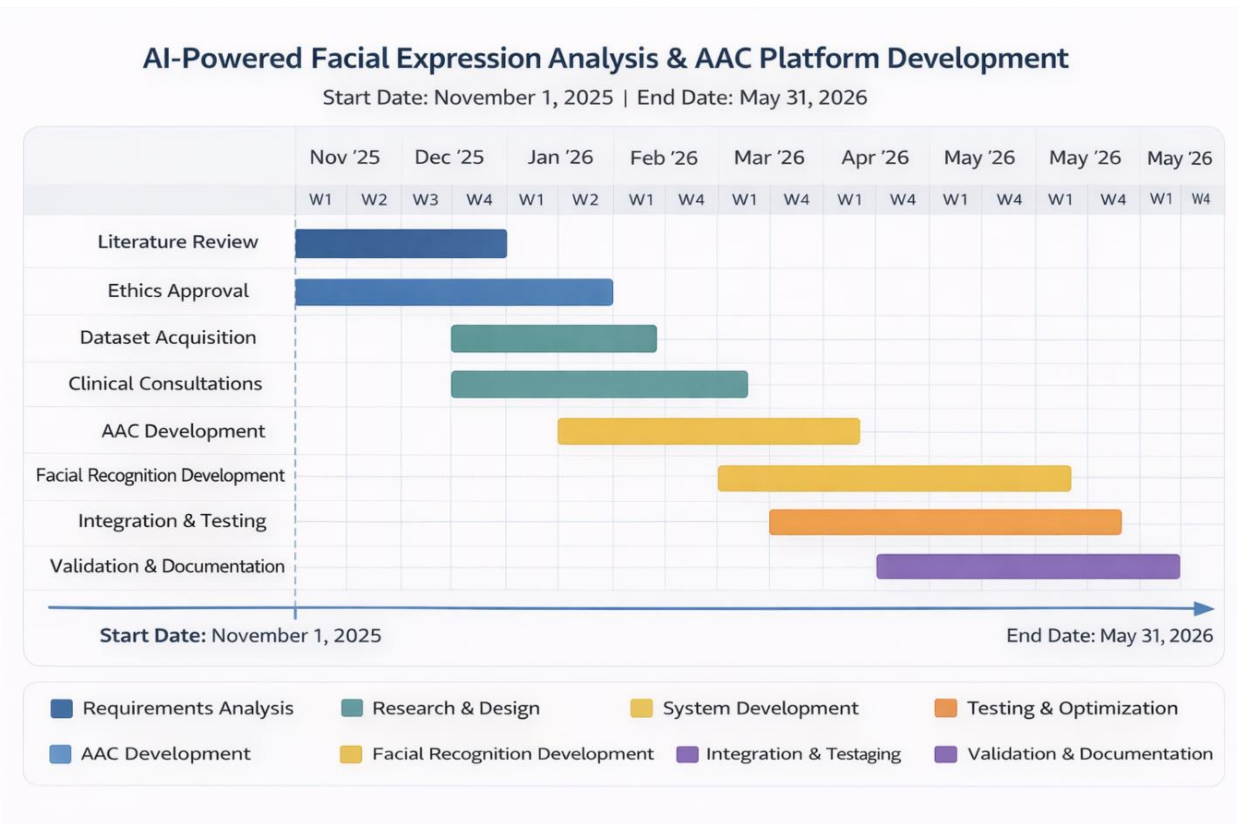
Suggested Starting Point

The project has already commenced with preliminary consultations involving approximately fifteen medical professionals affiliated with Karapitiya Teaching Hospital. These meetings were conducted to gather clinical insights, understand autism screening workflows, and define validation requirements for the proposed system. Based on this feedback, an initial prototype of the application was developed and presented to the medical team, who expressed positive interest in the solution and its potential social impact.

Following this initial phase, the author will proceed with a step-by-step development approach, beginning with user interface enhancement and functional improvements to the existing prototype. The system will be iteratively refined through continuous testing with children and parents in supervised environments. Feedback gathered from these user evaluations will be used to further improve usability, accessibility, and communication effectiveness. In parallel, publicly available datasets will be accessed for model training, and the technical development environment will be fully configured. This structured and iterative development process ensures that both research objectives and social impact goals are addressed effectively from the early stages of the project.

Gantt Chart

The following Gantt chart illustrates the complete project timeline from 1st November 2025 to 31st May 2026. The chart has been prepared using a project planning tool and represents the schedule for all major research, development, testing, and documentation activities of this project.



References

Millar, D.C., Light, J.C. and Schlosser, R.W., 2006. The impact of augmentative and alternative communication intervention on the speech production of individuals with developmental disabilities. *Journal of Speech, Language, and Hearing Research*, 49(2), pp.248–264.

Samad, M.D., Bobzien, J., Harrington, J., Iftekharuddin, K.M., Shashikumar, S. and Clifford, G.D., 2021. Automated facial expression recognition for children and adults with autism using computer vision. *IEEE Transactions on Affective Computing*, 12(4), pp.894–908.

World Health Organization, 2021. *Autism spectrum disorders*. Available at: https://www.who.int/news-room/fact-sheets/detail/autism-spectrum-disorders [Accessed January 2026].

Bibliography

Beukelman, D.R. and Mirenda, P., 2013. *Augmentative and alternative communication: Supporting children and adults with complex communication needs*. 4th ed. Baltimore: Paul H. Brookes Publishing.

Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M. and Adam, H., 2017. MobileNets: Efficient convolutional neural networks for mobile vision applications. arXiv preprint arXiv:1704.04861.

Light, J. and McNaughton, D., 2014. Communicative competence for individuals who require augmentative and alternative communication: A new definition for a new era of communication? *Augmentative and Alternative Communication*, 30(1), pp.1–18.

Flutter Documentation, 2024. Building adaptive apps. Available at: https://flutter.dev/docs

TensorFlow Lite, 2024. Guide to TensorFlow Lite. Available at: https://www.tensorflow.org/lite/guide

Google ML Kit Documentation, 2024. Face Detection API. Available at: https://developers.google.com/ml-kit/vision/face-detection

Kaggle, 2024. Autism Face Dataset. Available at: https://www.kaggle.com/datasets

National Institute of Health, 2024. Autism Research Database. Available at: https://www.nimh.nih.gov/research

Karolinska Institute, 2024. KDEF – Karolinska Directed Emotional Faces Dataset. Available at: https://www.kdef.se/